THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · MEDIUM LEVEL FULL-YEAR PAPER SET 2

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. Two numbers are in the ratio 3 : 5. If 4 is added to each, the numbers become 28 and 44. The original smaller number is: 1

(a) 28 **(b)** 24 **(c)** 48 **(d)** 20

2. A watch is sold for ₹ 1530 at a loss of 15%. Its cost price (in ₹) is: 1

(a) 1980 **(b)** 1620 **(c)** 1800 **(d)** 1759.5

3. A rectangle has a diagonal of 30 cm and one side 18 cm. Its perimeter (in cm) is: 1

(a) 96 **(b)** 42 **(c)** 86 **(d)** 84

4. 12 workers working 8 hours a day finish a job in 15 days. 10 workers working 6 hours a day will finish it in (days): 1

(a) 24 **(b)** 180 **(c)** 18 **(d)** 26

5. A pyramid has 7 vertices. The number of its edges is: 1

(a) 14 **(b)** 12 **(c)** 8 **(d)** 6

6. The mean of 59, 36, 46, 44, 50, 11 is 41. If 46 is replaced by 52, the new mean is: 1

(a) 49.67 **(b)** 47 **(c)** 42 **(d)** 41

7. $(x+2)(x+5)$ equals: 1

(a) $x^2+7x+10$ **(b)** $2x+7$ **(c)** $x^2+10x+7$ **(d)** x^2+7

8. A parallelogram has base 18 cm and height 10 cm. If the base is changed to 15 cm (same area, other side), the corresponding height (in cm) is: 1

(a) 12 **(b)** 13 **(c)** 15 **(d)** 165

9. The ones digit of 127^2 is: 1

(a) 1 **(b)** 4 **(c)** 9 **(d)** 7

10. If $3^x \times 3^4 = 3^{11}$, then x is: 1

(a) 44 **(b)** 7 **(c)** 15 **(d)** 9

11. XLIV + LXVII equals: 1

(a) CXXI **(b)** CXII **(c)** CXI **(d)** CI

12. The diagonals of a parallelogram ABCD meet at O. If AO $= (2x+3)$ cm and OC $= (3x-4)$ cm, then AC is: 1

(a) 36 cm **(b)** 34 cm **(c)** 17 cm **(d)** 7 cm

13. One step of the Kaprekar routine applied to 1895 gives: 1

(a) 8262 **(b)** 8163 **(c)** 8361 **(d)** 8280

14. The value of $63^2 - 37^2$ is: 1

(a) 10000 **(b)** 5338 **(c)** 2600 **(d)** 676

15. A mixture of 60 L has milk and water in the ratio 7 : 3. The quantity of water is: 1

(a) 42 L **(b)** 18 L **(c)** 13 L **(d)** 23 L

16. After a discount of 15%, a pair of shoes is sold for ₹ 1530. The marked price (in ₹) is: 1

(a) 1620 **(b)** 1800 **(c)** 1759.5 **(d)** 1980

17. The side of a rhombus is 30 cm and one diagonal is 36 cm. The other diagonal (in cm) is: 1

(a) 24 **(b)** 50 **(c)** 49 **(d)** 48

18. The ratio of boys to girls in a school is 7:8. If there are 1350 students, the number of girls is: 1

(a) 675 **(b)** 1335 **(c)** 630 **(d)** 720

19. All 8 faces of a polyhedron are triangles. The number of its edges is: 1

(a) 24 **(b)** 16 **(c)** 12 **(d)** 8

20. The median of 46, 96, 29, 15, 31, 39, 70, 52 is: 1

(a) 42.5 **(b)** 40 **(c)** 47.25 **(d)** 31

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. Two numbers are in the ratio 5 : 9 and their sum is 98. Find the numbers. 2

22. By selling an article for ₹ 1380, a trader gains 15%. Find the cost price. 2

23. An isosceles triangle has equal sides of 30 cm and base 36 cm. Find its height. 2

24. Two numbers are in the ratio 5:8 and their sum is 143. Find the numbers. 2

25. A cuboid is 15 cm long, 9 cm wide and 6 cm high. Find the sum of lengths of all its edges and the number of edges. 2

26. The mean marks of 15 boys is 60 and of 10 girls is 70. Find the mean mark of the whole class of 25. 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 40–60 words each.

27. A ladder 25 m long reaches a point 24 m up a wall. The foot is then pulled away so that it is 15 m from the wall. How far down does the top of the ladder slip? 3

28. 10 workers working 6 hours a day can finish a job in 24 days. In how many days can 12 workers working 8 hours a day finish it? 3

29. A cube of side 4 cm is painted all over and cut into 1 cm cubes. Find the number of cubes with 3

(i) exactly three painted faces
(ii) no painted face
(iii) at least one painted face.

30. The mean weight of 10 students is 42 kg and of another 15 students is 48 kg. Find the mean weight of all of them. 3

31. The sides of a triangle are $(2x + 3)$ cm, $(3x + 1)$ cm and $(4x + 5)$ cm. Find its perimeter, and its value when $x = 2$. 3

OR

A rectangular park is 40 m by 30 m. A path 2 m wide runs inside along its boundary. Find the area of the path. 3

32. Find the smallest number by which 12000 must be divided to make it a perfect cube. Find the cube root of the quotient. 3

OR

Evaluate $2^{-3} + 3^{-2}$ as a single fraction. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. A mixture of 60 litres has milk and water in the ratio 2:3. How much water must be added to make the ratio 2:4? 5

34. A prism and a pyramid both have a pentagonal base. Find the difference between the number of 5

(i) faces

(ii) vertices

(iii) edges of the prism and the pyramid.

35. The temperatures (°C) at 8 am, 10 am, 12 noon, 2 pm, 4 pm and 6 pm were 18, 22, 27, 30, 26, 21. 5

(i) Find the rise or fall between each consecutive pair.

(ii) Find the mean temperature.

(iii) In which interval was the change largest?

36. Expand and simplify $(x + 3)(x + 5) - x(x + 4)$. Find its value at $x = 2$. 5

OR

A figure is made of a rectangle 20 cm by 10 cm with a right triangle of base 20 cm and height 8 cm placed on one of its longer sides. Find the total area and the area of the triangle as a percentage of the total. 5

37. The volume of a cube is 1728 cm³. Find the side of the cube and its total surface area. 5

OR

The diameter of a cell is 2×10^{-5} cm. How many such cells placed side by side in a straight line make a length of 8×10^{1} cm? Give the answer in scientific notation. 5

38. Write the smallest and the largest 5-digit numbers in base 2 and find their decimal values. How many 5-digit numbers are there in base 2? 5

OR

The diagonals of a rectangular frame ABCD are AC $= (3x + 5)$ cm and BD $= (2x + 15)$ cm. Find x, the length of each diagonal, OA (O is the centre) and the length of wire needed to cover both diagonals. 5

END OF QUESTION PAPER — Total: 80 Marks

MINDSTORM